Allan Paiz

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Skills

- **Languages**: C++, Python, Java, Bash; Pandas, NumPy, Plotly, Matplotlib, Scikit-learn, JUnit
- **Tools**: LaTeX, Neovim, Git, GitHub, Linux, Excel, PowerPoint, Photoshop, Premiere Pro, Unreal Engine 5

Education

Bachelor of Science in Computer Science - University of South Carolina - Columbia, SC Aug 2023 - Dec 2026
3.7 GPA | Minor in Mathematics
Academic Honors: Fall 2023, Spring 2024, Spring 2025, Fall 2025
Relevant Coursework: Software Engineering, Analysis of Experimental Data in Python, Math Foundations of ML/Data Science, Optimization, Artificial Intelligence, Computer Architecture, Operating Systems, Linear Algebra, Technical Writing, Elementary Number Theory, Introduction to Computer Networks

Experience

Research Assistant - University of South Carolina - Columbia, SC Jan 2026 - Present

- I assist with research under Dr. Vignesh Narayanan using Python, C++, and Bash to run hardware tests with TurtleBot4 robots using ROS 2. On Linux, I collect experimental data for mobile ground robot control research. I develop and maintain simulation software in Gazebo for reproducible testing.

Advanced Data Collector - Pro Football Focus - Remote May 2024 - Present

- I collect NFL and NCAA football player data, focusing on speed, precision, and consistency. I have processed 38 games, identifying and recording over 210,000 data points. In 2024, I achieved an entry accuracy as high as 99.5%.

Research Assistant - NSF CISE Research Experience for Undergraduates - Columbia, SC May 2026 - Jul 2026

- I wrote a technical report proposing a ROS 2 perception architecture specifically for TurtleBot4 robots. After collecting camera and 2D LiDAR data, using Python I implemented sensor fusion and computer vision algorithms for object detection to demonstrate the pipeline.

US Marine - United States Marine Corps - Camp Pendleton, CA May 2018 - Oct 2022

- I led and trained teams in technical environments under high pressure. I served as a Sapper Leaders Course instructor responsible for training and evaluating platoons. I operated the ENFIRE digital reconnaissance system to collect, analyze, and communicate geospatial and engineering data.

Technical Designer - Production Resource Group - New Windsor, NY Jun 2015 - Dec 2016

- I collaborated with stakeholders to engineer solutions for entertainment scenery and machinery projects, meeting strict safety, functional, budget, and timeline constraints. I used AutoCAD for 2D and 3D modeling, drafted technical drawings for manufacturing, and created assembly instructions and bills of materials.

Projects

Wash Your Rice - Unreal Engine Game - Group Project August 2025 - May 2026

- I worked with a team to develop a horror puzzle game in Unreal Engine 5 using C++. We followed a game development lifecycle and utilized GitHub for collaboration and version control. [Capstone Page]

Construction Permit Delay Risk Prediction - ML/Data Science Course Project Jan 2026 - May 2026

- Using Python, I processed open source permit data to identify construction projects at high risk of delay. Using Scikit-learn, I built and compared a kernel SVM with neural network classifiers to predict which permits were at high risk of delay.

Optimal Control for Obstacle-Avoidant Navigation - Optimization Course Project Jan 2026 - May 2026

- I assisted in developing a framework for obstacle avoidance trajectory planning for wheeled mobile robots. We formulated the task as a constrained nonlinear optimization problem and fed the resulting trajectory into a neural network for event-triggered optimal control.

Defensive Stopping Power - 2024 NFL Big Data Bowl - Kaggle Competition Oct 2023 - Jan 2024

- I developed a feedforward neural network in Python to detect contact between ball-carriers and defenders using NFL player tracking data. I created a novel metric, Yards Allowed After Contact, to evaluate defensive performance

for player scouting and game analysis. [Project Link] [Finalists Link]

TuneUp by JETA - Software Engineering Course - Team Project Jan 2025 - Apr 2025

- I worked with a team to develop a music application written in Java. We followed an agile, iterative development process using design patterns, unit testing, and object orientation. I created documentation and system design diagrams and used GitHub for version control. [GitHub Link] [Demo Video]

Awards

NFL Big Data Bowl - Runner Up 2024
My submission was selected as a finalist, earning a runner up prize of $5,000. [NFL Announcement]

Marine Corps Good Conduct Medal 2021
Outstanding performance & conduct during three years of continuous active enlisted service in the U.S. Marine Corps.

Certificate of Commendation 2021
Outstanding performance of duty while serving as a seminar leader for Lance Corporal Leadership and Ethics Seminar.